

pass 1 · teacher · stop-grad

Full paragraphs $\mathcal{P}$
 N clean strings

pass 2 · student · trainable

Kept paragraphs $\mathcal{P}_{\text{kept}}$
 $M < N$ · paragraph dropout 30%

pass 3 · MLM · trainable

Token-masked paragraphs
 N strings · MLM 15% (BERT-style 80/10/10)

DistilRoBERTa — *shared encoder* (one `RobertaForMaskedLM` instance)
 three forward passes per step · gradient checkpointing · weights identical across passes

$\mathbf{H}_T \in \mathbb{R}^{N \times 768}$
 CLS per paragraph

$\mathbf{H}_S \in \mathbb{R}^{M \times 768}$
 CLS per kept paragraph

$\mathbf{H}_M \in \mathbb{R}^{N \times L \times 768}$
 hidden per token

Mean Pooling
 $\frac{1}{N} \sum_{i=1}^N$

AttentionPooling
 learnable query + pos.

LM Head
 linear $768 \rightarrow V$ · *inside the shared RobertaForMaskedLM*

$\bar{\mathbf{H}}_T \in \mathbb{R}^{768}$
 teacher target (detach)

$\mathcal{L}_{\text{para-pred}}$
 $1 - \cos(\mathbf{z}_{\text{partial}}, \bar{\mathbf{H}}_T)$

$\mathbf{z}_{\text{partial}} \in \mathbb{R}^{768}$
 student partial pool

Logits $\in \mathbb{R}^{N \times L \times V}$
 per-token vocab dist.

$\mathcal{L}_{\text{mlm}}$
 cross-entropy on <mask>

$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{mlm}} + (1 - \alpha) \mathcal{L}_{\text{para-pred}}$
 $\alpha = 0.5$ — jointly trains encoder + AttentionPooling

— — — stop-gradient pass — — — trainable pass — — — loss signal — — — shared encoder (single block above)